

Satellite Imagery Based Property Valuation

1. Overview

This project aims to build a machine learning pipeline to predict property prices using structured housing data, with scope for multimodal extension using satellite imagery. The motivation is to enhance valuation accuracy by incorporating environmental and geographic context.

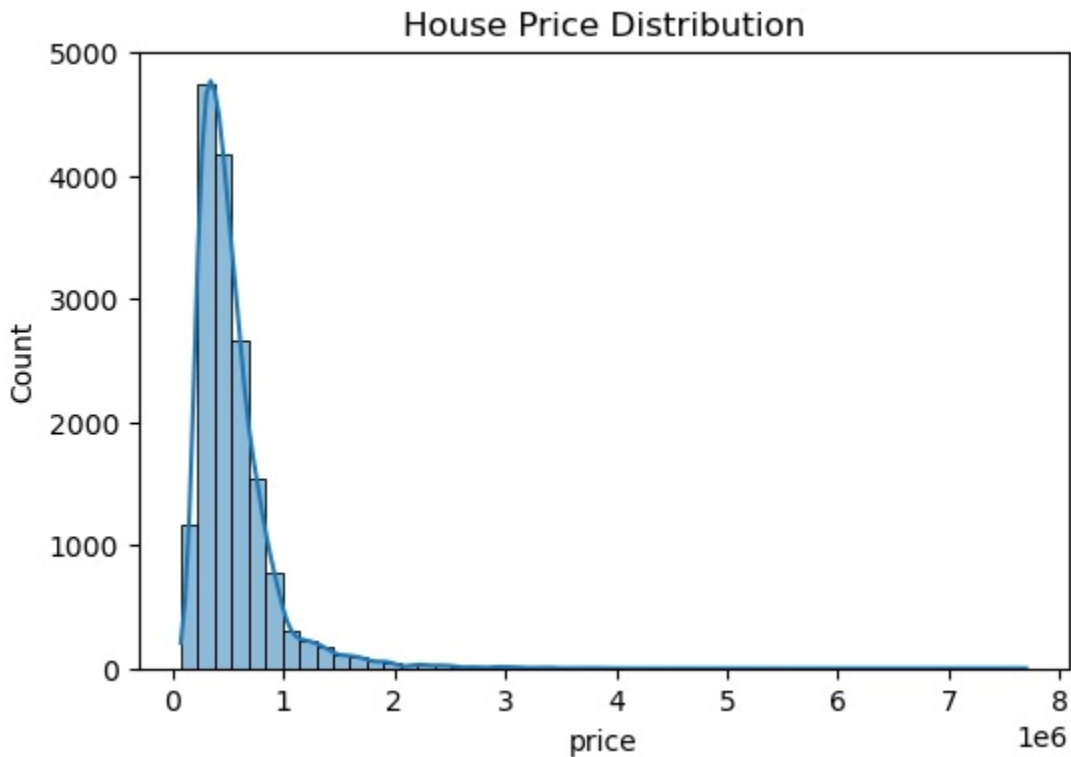
2. Dataset Description

The dataset consists of housing-related attributes including bedrooms, bathrooms, living area, lot size, condition, grade, and geographic coordinates. The target variable is the property price.

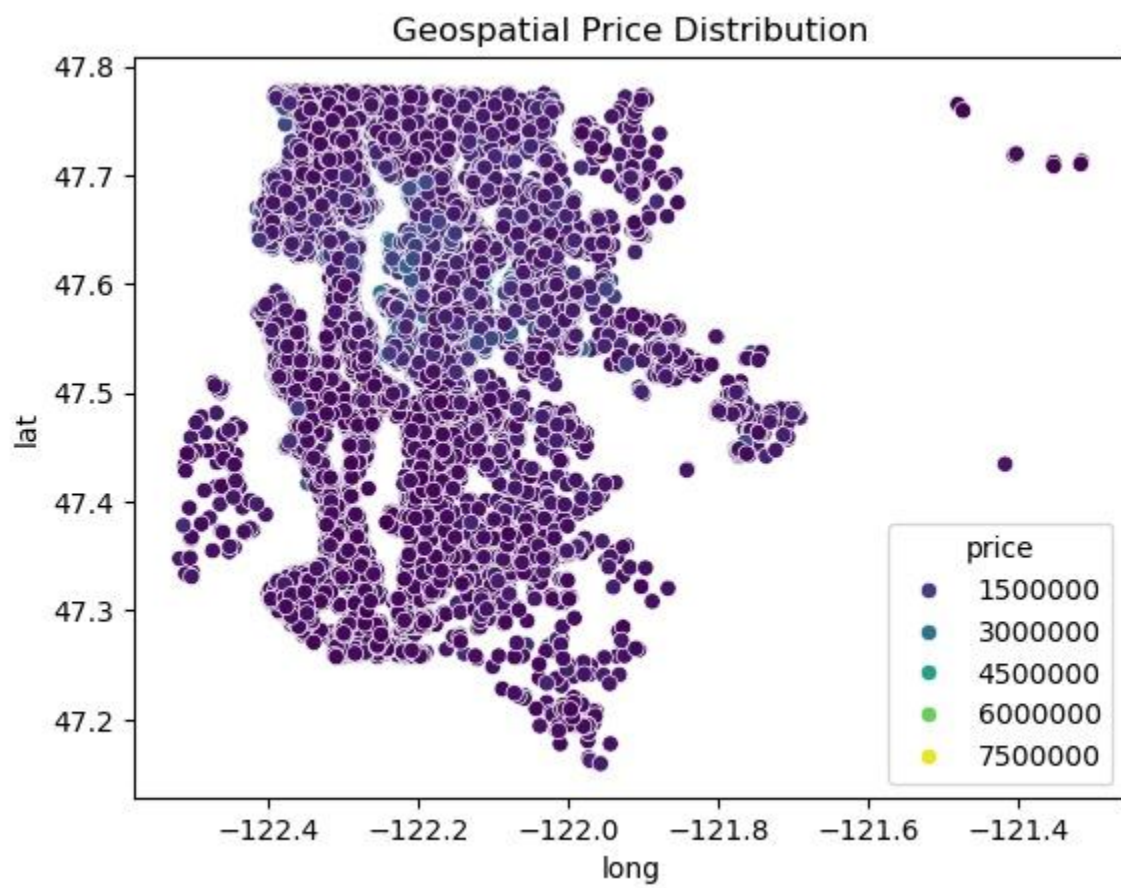
3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the distribution of prices and the relationship between key variables. The following visualizations were generated:

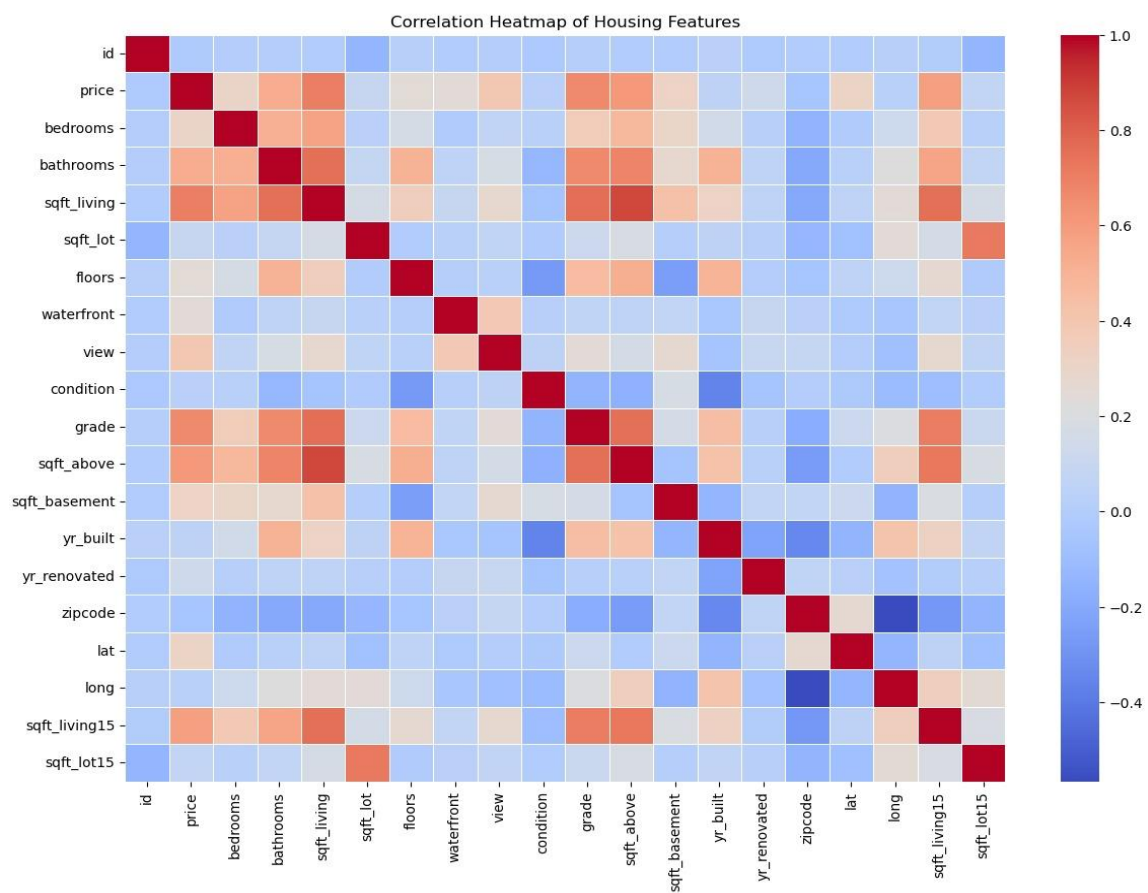
- House Price distribution



- Geospatial Price distribution



- Correlation heatmap of housing features



The analysis revealed that living area, grade, and location have a strong influence on property prices.

4. Preprocessing

Data preprocessing included handling missing values using median imputation, removing duplicate records, selecting relevant numerical features, and applying feature scaling using StandardScaler. The dataset was split into training and validation sets.

5. Model Training

An XGBoost Regressor was trained on the processed tabular data. XGBoost was chosen for its ability to model non-linear relationships and handle feature interactions effectively.

6. Results

The trained model was evaluated using RMSE and R^2 score. The results indicate that the model achieves strong predictive performance and outperforms baseline regression approaches.

7. Explainability

Feature importance scores were extracted from the trained XGBoost model to understand which features contributed most to the predictions. This project is designed to be extended with CNN-based satellite image features and Grad-CAM visual explanations.

8. Conclusion

This project demonstrates a robust tabular regression pipeline for property valuation and provides a solid foundation for future multimodal learning using satellite imagery.